# Year 8 Mathematics Test Chapters 1 - 4 End Term 1

## Resource free Section

**Instructions:** 1. Answer all questions

Baldivis Secondary College

Time: 10 mins Name: Mark Inkey. /10 marks

1 Substitute $x = 3$ into $2x + 1$ and then evaluate.

$2(3)+1 = 7$ ✓

2 $4 + {}^{-}6$

$-2$ ✓

3 $7 - {}^{-}4$

$11$ ✓

4 ${}^{-}5 \times {}^{-}3$

$15$ ✓

5 Simplify: $6n - 3n$.

$3n$ ✓

6 A fair 6-sided die is tossed, what is Pr (3)?

$Pr(3) = \frac{1}{6}$ ✓

7 Distribute (*hint: expand*): 2(a + 2)

$2a + 4$ ✓

8 Factorise (*hint: put into brackets*): 2x + 8.

$2(x+4)$ ✓

9 Pr(winning) = 0.6, what is Pr(not winning)?

$0.4$ ✓

10 Simplify (*hint: index laws*): $x^5 \times x^2$

$x^7$ ✓

# Year 8 Mathematics Test Chapters 1 - 4 End Term 1

**Instructions:** 1. Answer all questions
2. Calculators permitted 3. One Page of notes, one sided

Time: 40 mins Name: ______________________ /50 marks

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## Question 1 (7 marks)

**a)** Write each of the following in index form:

**i)** $2 \times 2 \times 2$

$2^3$ ✓

**ii)** $d \times d \times d \times d \times d \times d$

$d^6$ ✓

**b)** Simplify and write the following in index form:

**i)** $3^2 \times 3^4$

$3^6$ ✓

**ii)** $3^4 \div 3^3$

$3$ ✓

**iii)** $(4^2)^3$

$4^6$ ✓

**iv)** $3^5 \times (3^2)^3$

$3^5 \times 3^6 = 3^{11}$ ✓
✓

## Question 2 (3 marks)

For the following pairs of numbers, place < (less than) or > (greater than) between them to make a true statement.

**i)** $5 > {}^-1$ ✓ **ii)** ${}^-3 < 3$ ✓ **iii)** ${}^-4 < {}^-3$ ✓

**Question 3 (23 marks)**

**a)** Write an algebraic expression for each of the following:

**i)** the sum of a and b.

$a+b$ ✓

**ii)** h is decreased by 2x.

$h-2x$ ✓

**b)** Simplify the following expressions:

**i)** $2a + 7a$

$9a$ ✓

**ii)** $5b - 3b$

$2b$ ✓

**iii)** $6c \div 3$

$2c$ ✓

**iv)** $3 \times 3d$

$9d$ ✓

**c)** Expand and then simplify if necessary, each of the following expressions:

**i)** $3(b + 5)$

$3b+15$ ✓

**ii)** $2(a + 3) - 5$

$2a+6-5$ ✓ $= 2a+1$ ✓

**iii)** $-2(4y - 7)$

✓

$-8y+14$ ✓

or $14-8y$

**iv)** $3x(x + 2) + 2(x + 3) - 7$

$3x^2+6x+2x+6-7$ ✓

$= 3x^2+8x-1$ ✓

**d)** Factorise each of the following expressions:

**i)** $3d + 6$

$3(d+2)$ ✓

**ii)** $3xy + 21x$

$3x(y+7)$ ✓ ✓

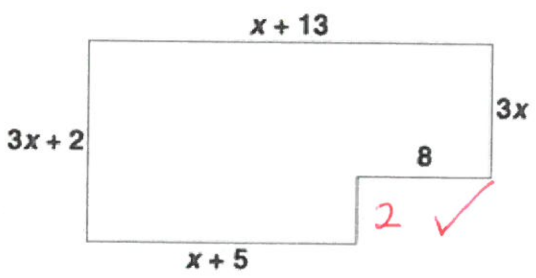
**iii)** $2x^2 - 6x$

$2x(x-3)$ ✓ ✓

**iv)** $5x - 15xy + 45$

$5(x - 3xy + 9)$ ✓ ✓

**e)** What is the perimeter of the shape?




$x + 13 + 3x + 8 + 2 + x + 5 + 3x + 2$ ✓

$= 8x + 30$ ✓

## Question 4 (14 marks)

**a)** A six-sided die is thrown. What is the probability of each of the following?

**i)** 2?

$Pr(2) = \frac{1}{6}$ ✓

**ii)** even?

$Pr(\text{even}) = \frac{3}{6} = \frac{1}{2}$ ✓

**iii)** <3?

$Pr(<3) = \frac{2}{6} = \frac{1}{3}$ ✓

**iv)** a number divisible by 3?

$Pr(\div 3) = \frac{2}{6} = \frac{1}{3}$ ✓

**b)** There are 500 tickets in a raffle raising money for the Junior basketball club. If you bought ten tickets, what is the probability of:

**i)** winning the raffle?

$Pr(\text{win}) = \frac{10}{500} = \frac{1}{50}$ ✓

**ii)** not winning the raffle?

$Pr(\text{not win}) = \frac{490}{500} = \frac{49}{50}$ ✓

**c)** A bag contains ten red marbles, ten blue marbles, and five green marbles. What is the probability that the first marble taken from the bag is:

**i)** red?

$Pr(\text{red}) = \frac{10}{25} = \frac{2}{5}$ ✓

**ii)** not red?

$Pr(\text{red}') = \frac{15}{25} = \frac{3}{5}$ ✓

**iii)** not green?

$Pr(\text{green}') = \frac{20}{25} = \frac{4}{5}$ ✓

**d)** The letters CAT can be rearranged in six ways:

**i)** Show the ways they can be rearranged (hint: they do not need to make words).

CAT ACT TAC
CTA ATC TCA

✓ ✓ ✓

**ii)** What is the probability that the arrangement will start with the letter A?

$Pr(A) = \frac{2}{6} = \frac{1}{3}$ ✓

**iii)** What is the probability that the letters C and T are next to each other?

$Pr(C \,\&\, T) = \frac{4}{6} = \frac{2}{3}$ ✓

## Question 5 (3 marks)

A coin was tossed 60 times. The results are shown in the table below:

| | Heads | Tails |
|---|---|---|
| Number of times the result was obtained | 21 | 39 |

**i)** Using these results, what is the probability of obtaining a tail?

$Pr(\text{Tail}) = \frac{39}{60} = \frac{13}{20}$ ✓

**ii)** What is the Theoretical Probability of obtaining a tail?

Theoretical $Pr(\text{Tail}) = \frac{1}{2}$ ✓

**iii)** Are these probabilities the same? Why/Why not?

No because theoretical probability . . .

any reasonable explanation.

(must have an explanation) ✓

**END OF TEST**